Applying Evolutionary Systems Engineering to Rural Energy Roadmaps: A Socio-Technical Perspective

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***Abstract*—Sub-saharan *africa is characterised by sparsely populated areas, small communities limited infrastructure, and signficant self reliance. These, nano-communities, typically consist of 3-20 households, which are too small to justify typical energy infrastructure (posing a challenge to local municipalities) and have modest energy needs, yet serve society in cultural, agricultural, and tourism. This presents an imminent need to develop energy solutions tailored for nano-communities. This paper roadmaps the development of nano-community energy solutions*. Systems engineering is used to roadmap the solution using an iterative development approach. Early stages of the roadmap are demonstrated by implementation of energy access solution in communities and were positively received. Interestingly, the challenge that emerged were the rapidly changing requirements in the form of increased energy appetite from the community, which is distinct from typical socio-technical system behavior. Through this work, two things become apparent. Firstly by seperating electrical and thermal energy needs, we are able to develop cost effecitive Solar home systems to meet electrical needs, while meeting thermal needs from wood-fired combustion. This makes decentralized energy more accessible. Secondly, socio-technical energy road-mapping should consider cultural norms, like woodfire cooking, and increasing energy appetite.**

Keywords—Rural Electrification, Solar Home Systems, Frugal Innovation, Iterative Development, Technology Roadmapping

I. Introduction

Sub-Saharan Africa (SSA) is characterized by long distances, sparse population density, and geographically isolated communities. These conditions make decentralized energy access solutions particularly attractive. However, previous electrification efforts have often achieved limited success due to overly ambitious planning, insufficient community engagement, and misalignment with local socio-economic conditions.

A staggered, iterative implementation approach that incorporates continuous stakeholder feedback is essential to reduce the risk of failure and accommodate evolving “energy appetite.” International carbon border tax is causing industrial sectors in the region prioritize renewable energy for economic sustainability [1], but this mechanism excludes rural populations as they are unaffected by tax, leaving little incentive for the renewable this transition.

In South Africa, the government continues to prioritize centralized grid expansion despite evidence favoring decentralized systems for remote areas. Approximately USD 24 billion has been committed to expanding the national transmission network [2], including an estimated 14,000 km of new infrastructure [3]. While South Africa reports a high national electrification rate of 94% [4], the remaining unelectrified populations are the most difficult and costly to reach via traditional grid extension. This strategy persists despite findings that Solar Home Systems (SHSs) can provide electricity at a lower cost in sparsely populated regions [5]. In contrast, countries like Rwanda have successfully accelerated access by combining grid and off-grid solutions, increasing electricity access from 10% in 2010 to 43% in 2018—with nearly 48% of that growth coming from off-grid systems [6]. During the same period, South Africa’s grid-first approach saw access increase only marginally from 76.9% to 79.9% [7]

The barriers to SHS adoption are primarily socio-technical rather than purely technological [8]. For example, South Africa’s 1999 SHS concessions—a public-private partnership—failed due to poor planning, low-quality components, and a lack of adherence to standards, resulting in the systems being perceived as inferior to grid electricity [9]. Conversely, in Kenya, where off-grid connections are often preferred by those with a higher willingness to pay [10], adoption has been more robust. These contrasting cases highlight that implementation details like component quality, regulation, and public perception are more critical to adoption than the technology itself.

A. Capacity requirements of rural SHSs in SSA and appliance tier transition

A primary reason for mixed results in SHS implementation is the failure to recognize that rural electrification is a "dynamically complex" problem: by the time a technical solution is deployed, the community’s requirements have already changed. While rural settlements often begin with modest energy needs of approximately 100W for lighting and basic appliances [11], the introduction of power triggers a rapid increase in energy appetite [12].

This is evidenced by "appliance tier transition." As households gain access, they transition quickly from basic lighting (Tier 1) to entertainment (Tier 2) and medium-intensity appliances like blenders or kettles (Tier 3) [12], [13]. However, transition to Tiers 4 and 5 (refrigeration and

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heating) is often stalled by the high costs of both appliances and energy. Furthermore, while social outcomes—such as increased income generation, education, and health benefits from reduced smoke inhalation—are significant [14], the high energy intensity of thermal loads (cooking and space heating) remains a challenge for SHS configurations as it significantly increases the system cost. Due to the complexities of rural finances, mobile payment models like Pay-As-You-Go (PAYG) have proven successful in bridging the gap [15].

B. Nano-communities

This paper introduces the term “nano-communities” to describe geographically isolated clusters of 3–20 households commonly found in remote areas. These communities are often located far from existing infrastructure, making grid extension economically unviable. In the South African context, such communities frequently emerge around agricultural activities and seasonal labor patterns.

A previous study [16] identified critical user requirements through community interviews, specifically prioritizing lighting, smartphone charging, and cooking. That research revealed that households were spending approximately R200 per month on hazardous candles, an amount that could be redirected toward sustainable energy solutions. This work deployed an SHS, but was limited to a signle implementaion. While the previous work successfully deployed a standalone Solar Home System (SHS), it was limited to a single implementation and could not address cooking, as meeting energy-intensive thermal loads with basic PV systems remains a significant challenge. Furthermore, the earlier study did not take into account the "increased energy appetite" identified in literature, which highlights that rural electrification is a "dynamically complex problem" where requirements evolve rapidly as soon as initial modest needs are satisfied.

C. Proposed Evolutionary Methodology

To further this development, this study proposes an Evolutionary Socio-Technical Systems Engineering approach based on the Vee model [17]. By utilizing technology demonstrators—prototype subsystems used to test high-risk technologies like IoT monitoring and thermal batteries in controlled or limited real-world deployments prior to full-scale implementation—we can navigate technical risks before full deployment. This roadmap transitions from basic DC Solar PV SHS [16] toward future-states that enable peer-to-peer energy sharing [18]. This methodology ensures that the energy solution remains culturally appropriate and sustainable by evolving alongside the community it serves.

METHODOLOGY

This study integrates multiple complementary frameworks into a unified methodology. The Vee model serves as the primary systems engineering backbone, structuring the development lifecycle from requirements to validation. The Vee model is modified with pre and post activities to stakeholder-led system design. While Evolutionary Systems Engineering (ESE) uses the Vee-model at each iteration but introduces iterative adaptation to accommodate dynamically changing requirements enables continuous incorporation of community feedback.

The Socio-Technical Systems (STS) perspective provides the analytical lens to capture the interaction between human and technical elements. Together, these elements form a modified Vee model tailored for rural energy system development.

The design and implementation of energy systems for rural nano-communities necessitate a methodology that accounts for the complex interplay between social norms and technical requirements. This study conceptualizes the problem through the lens of Socio-Technical Systems (STS), defined as systems operating at the functional intersection of social and technical dimensions [19]. Given the dynamic nature of energy demand in these contexts, a static design approach is insufficient. Consequently, this research adopts the stakeholder-led system design framework proposed by [20], which utilizes regular community reviews to mitigate the risk of technical obsolescence caused by shifting "energy appetites."

To manage evolving requirements that cannot be fully defined upfront, an Evolutionary Systems Engineering approach is employed [17]. This methodology utilizes an iterative visualization of the Vee Model [21] to align technical development with community feedback loops.

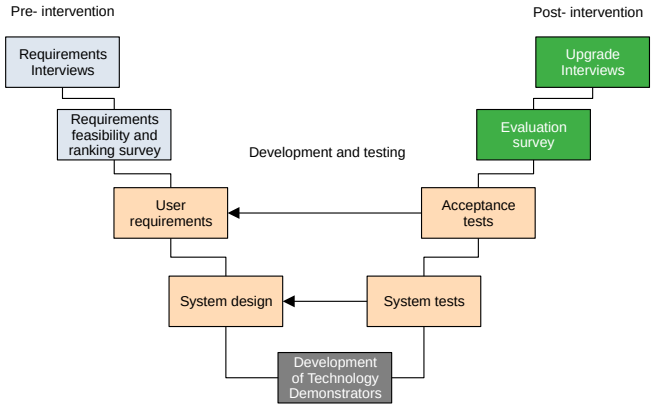


Fig. 1. Figure 1: Modified Vee-model containing the pre-intervention, conventional development and testing, and post-intervention.

As illustrated in Figure 1, the proposed methodology integrates stakeholder engagement into three distinct phases of the lifecycle:

1. Pre-intervention: Initial stakeholder needs are ascertained through an semi-structured interview process in this exploration phase. During this stage, fundamental requirements—typically lighting, cooking, and basic income-generating activities—become apparent. To ensure technical and economic viability, a feasibility study is conducted to eliminate impractical solutions. For example, high-power thermal loads such as electric cooking are filtered out at this stage, as their inclusion would necessitate prohibitively large battery capacities for a standard Solar Home System (SHS). Finally, a quantitative survey is administered to rank the remaining requirements based on community priority, ensuring the design is community-led.
2. Development and testing: The system is developed according to the established Vee model [22]. This phase focuses on translating the prioritized requirements into technical specifications, subsystem design, and prototype integration, while utilizing technology demonstrators to mitigate risks associated with untested components.
3. Post-intervention and Long-term Evaluation: Following deployment, a short-term survey is conducted to evaluate the system’s immediate efficacy in meeting the established requirements. To

account for the "dynamic complexity" of energy access, the community is revisited after an extended operational period. This long-term evaluation assesses shifts in energy behavior and identifies emergent needs—such as the transition from basic lighting to digital connectivity—which subsequently inform the next iteration of the technological roadmap.

This technical evolution is managed through Technology Roadmapping, defined as the process of planning a system's progression toward a specific end-state (MITRE, 2011), by integrating future technologies. To navigate the risks inherent in untested technologies, the roadmap utilizes Technology Demonstrators [23]. These demonstrators serve as experimental subsystems that validate feasibility in a controlled environment before full-scale integration. This iterative design ensures that the system remains adaptive, transitioning from basic DC prototypes toward a resilient, culturally appropriate energy infrastructure.

II. Resulting Roadmap

This section presents the resulting energy development roadmap derived from the proposed methodology. The roadmap translates stakeholder requirements and technical constraints into a phased implementation strategy, explicitly accounting for evolving energy demand and socio-technical adaptation.

A. Roadmap

The resulting roadmap, shown below, utilizes evolutionary systems engineering to electrify nano-communities, providing a framework to navigate the "dynamically complex" nature of rural energy demand. To address the community's evolving needs, development is staged across three iterations. Phase 1 focuses on immediate relief through cost-effective, standalone SHS for lighting and mobile charging. Phase 2 introduces sustainable wood-fired rocket stoves for cooking and heating alongside IoT demand profiling to track evolving usage patterns. Phase 3 represents the visionary end-state of Direct-DC micro-grids enabling peer-to-peer energy trading and unified information systems.

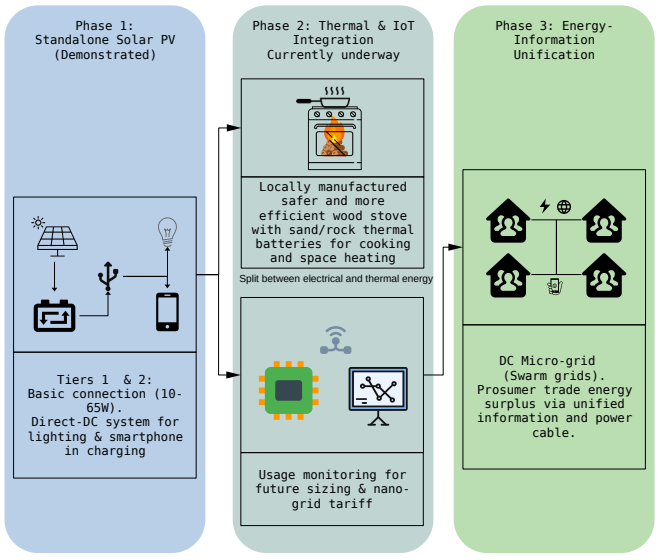


Fig. 2. Evolutionary roadmap for rural energy resilience, illustrating the phased transition from standalone solar-thermal demonstrators (Phases 1 & 2) to interconnected Direct-DC micro-grids and peer-to-peer energy trading (Phase 3).

B. Phase 1: Standalone Solar PV for lighting and smartphone charging

Phase 1 is focused on developing a cost-effective, standalone solar PV device to provide essential lighting and mobile charging. Given the communities' modest electrical requirements, the design utilized a direct DC system to eliminate the cost, complexity, and effeciency loss of an inverter. While stakeholder engagement identified cooking as a high priority, it was postponed during this phase due to technical and economic infeasibility. Post-implementation feedback confirmed the prototype successfully met initial user needs; however, it also revealed a rapidly increasing "energy appetite," with residents expressing new requirements such as internet access (e.g., via local Wi-Fi networks).

To provide a rigorous foundation for the "frugal innovation" claims, the technical capacity and economic viability of the Phase 1 prototype are detailed below:

The system was engineered to sustain a calculated daily power consumption of 47 Wh, providing approximately four hours of combined lighting and mobile charging.

TABLE I. Technical Specifications: Phase 1 SHS Prototype.

Feature	Specification	
	Minimum	**Realized**
PV Panel Wattage	13.53 W	100 W
Battery Capacity	5.74 Ah	7 Ah LiFePO4
System Voltage	5V DC	12 V (DC-only system, no inverter)
Charging Ports	1 x USB-Type A ports	2 x USB-Type A ports
Lighting Output	1 x Rechargeable LED Lanterns (USB-powered)	2 x Rechargeable LED Lanterns (USB-powered)

Cost Analysis and Frugal Innovation Validation
Field observations and prior studies indicate that the Phase 1 system is a social innovation because it targets a price point aligned with current user spending on hazardous alternatives like candles.

- Phase 1 Component Cost: The total cost of the prototypes ranged from R1,118.67 to R1,539.27.
- Economic Impact: Households in the target communities reported spending approximately R200 per month on candles. By diverting this existing expenditure toward a "rent-to-own" model for the PV system, users achieve an eight-month payback period.
- Thermal Feasibility vs. Electrical Needs: A critical finding in the feasibility study was that addressing thermal loads (cooking and heating) via solar PV was "infeasible". Thermal loads are inherently "energy intensive", and attempting to meet them with solar power would require prohibitively large battery capacities and increased device complexity that would "defeat the low-cost objective" of the design.
- The Roadmap Justification: This technical gap is what necessitated Phase 2 of your roadmap—the development of the woodfired rocket stove—to address thermal needs using sustainable biomass rather than expensive, oversized electrical storage.

C. Phase 2: Thermal Systems and IoT Demand Profiling

Following the finding in Phase 1 that solar-powered cooking was technically and economically infeasible for a low-cost system, Phase 2 introduced a wood-fired rocket stove

to address thermal needs. This phase also initiated the development of an IoT monitoring module to track the community's evolving energy consumption patterns.

1. Sustainable Wood-Fired Rocket Stove

The rocket stove was designed for the dual purpose of household cooking and indoor space heating, prioritizing environmental sustainability and local manufacturability [24]. To ensure safety and efficiency, the design utilized "thermal battery" layers composed of sand and rock, which are locally abundant and non-toxic.

Technical Performance and ISO Testing Laboratory testing conducted under BS ISO 19867-1:2018 (efficiency) and BS EN ISO 13732-1:2008 (safety) standards validated the prototype's superiority over traditional open fires.

TABLE II. TECHNICAL PERFORMANCE COMPARISON OF ROCKET STOVE AGAINST OPEN FIRE.

Thermal Efficiency
Traditional-use Conditions: The rocket stove achieved a thermal efficiency of 5.98%, more than double the 2.92% recorded for an open fire. Thermal efficiency was measured according to standardized water boiling test protocols defined in BS ISO 19867-1:2018.
Equal-height Configuration: Efficiency reached 7.44%, a 3.4-fold improvement over the open fire's 2.15%.
Cooking Power: The stove delivered 584.6 W of power, approximately 63% higher than the open fire's 357.5 W.
Safety Metrics
External Shell Temperature: The insulation strategy reduced the external shell temperature from a baseline of 460 °C to 74 °C. This performance nearly meets the ISO 13732-1 burn threshold of 73 °C for brief accidental contact.
Handle Temperatures: Testing confirmed that handles remained at <35 °C, well within the 51 °C safety limit for manual operation.
Thermal Inertia and Space Heating
The sand insulation provided 3.1 MJ of sensible heat storage.
This enabled a modest space-heating output of ~145 W for up to six hours after the fire subsided, sufficient to regulate the temperature of a small (9 m^2) room.

The effeciency of the stove achieved is 7,4% which is significantly higher than open fires, but is low compared to typical rocket stoves (estimated 20-40%). This is likely due to the low temperatures enforeced by ISO-13732 and should be investigate further.

2. IoT Demand and Generation Profiling

A primary challenge in rural electrification is the "increased energy appetite" that emerges once basic needs are met. To address this "dynamically complex problem," an IoT measuring and monitoring module was developed to meter the electrical power flow entering and leaving the Phase 1 solar PV system.

Closing the Data Gap While demand synthesizers exist for industrial systems, there is a critical lack of data regarding rural residential energy usage. The IoT module serves three vital functions:

- Capacity Planning: Accurate usage patterns aid in properly sizing systems and detecting when capacity upgrades are required.
- Tariffing and Monetization: Metering capabilities allow for user tariffing, creating potential business opportunities by monetizing the nano-energy grid.
- Socio-Technical Integration: By quantifying behavioral changes and growing demand, the module provides the technical foundation for the Direct-DC micro-grids and peer-to-peer energy trading envisioned for Phase 3.

D. Phase 3: Visionary Integration and Peer-to-Peer Energy Trading

Phase 3 represents the speculatory future state of the rural energy roadmap, transitioning from isolated standalone units to interconnected Direct-DC micro-grids. In this decentralized architecture, households evolve from mere consumers into prosumers who generate their own power and share surplus energy with neighbors through efficient DC interconnections. The compounding effect of modest energy needs, energy sharing, direct-DC grids contribute toward cost effective community-wide energy resilience as the collective "energy appetite" continues to expand.

The technological foundation of this phase involves a profound unification of energy and information systems, where power distribution and data communication occur over a single, consolidated infrastructure. This convergence enables robust connectivity across the settlement, supporting the real-time data requirements of the IoT demand profiling modules initiated in Phase 2. By integrating communication directly into the physical energy grid, the community establishes the necessary hardware to facilitate smarter, more responsive resource management.

To ensure long-term sustainability and financial independence, Phase 3 introduces local tariffing models and agent-based trading platforms. These automated systems allow for the fair exchange of surplus power between prosumers, potentially giving rise to a "zero marginal energy" economy where additional energy can be utilized to drive technological development at a near-zero cost. A primary socio-technical priority during this stage is ensuring that the community can maintain the system internally to prevent over-dependence on external entities and to protect human well-being.

Finally, the roadmap envisions a complete integration of thermal and electrical systems through the comprehensive use of IoT technology. Sensors incorporated into the domestic environment will monitor critical factors such as indoor air quality, the formation of hazardous gases, and general habitability. By unifying these thermal safety metrics with the electrical grid's monitoring platform, residents gain a holistic view of their domestic energy environment, ensuring both energy security and household health.

III. CONCLUSION

In conclusion, this novel approach demonstrates a solution that is cost-effective, culturally respective, and sustainable. By moving beyond simple grid-extension models, the roadmap provides near-future predictions of what possibilities become available as a community’s "energy appetite" evolves—from basic lighting to peer-to-peer energy trading and unified information systems.

While the Socio-technical Vee integration into technology roadmaps provides a general framework for engineering, the resulting roadmaps themselves do not generalize well across different regions. This lack of broad scalability is a necessary

tradeoff when dealing with "socio-technical wicked problems," where the specific cultural, spatial, and economic realities of a nano-community must dictate the technical design.

Ultimately, the success of such interventions relies on recognizing that the issues faced by rural residents—such as the danger of snakes or the high cost of traveling to charge a phone—are often difficult for most researchers to relate to. Therefore, constant communication and validation are required to ensure the technology remains "socially meaningful" rather than just technically sound. The following table illustrates how specific user feedback from the communities has directly shaped the roadmap's technical and social implications.

TABLE III. USER FEEDBACK AND ROADMAP INTEGRATION.

User Feedback	*Socio-Technical Category*	*Roadmap Implication*
"Light interfaces broke while SHS is still working"	Technical Reliability & Component Quality	Phase 2: Prioritize ruggedized or repairable interfaces to prevent functional obsolescence and ensure system longevity.
"With a candle, the snakes see me, with a light I can see the snakes"	Social Outcomes (Safety & Health)	Phases 1 & 2: Validate lighting as a critical safety requirement for "human well-being," rather than just a luxury or entertainment feature.
"Save money on candles/charging"	Economic Sustainability & Frugal Innovation	Phase 3: Leverage the high willingness to pay for systems that offset the high recurring costs of candles and kerosene (~R200/month) to fund local micro-grids.
"Don't have to walk to the shop"	Temporal Efficiency (Time Poverty)	General: Enhances household productivity and well-being by reducing the labor-intensive burden of seeking energy outside the home.

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